Lab Sheet 3: SQL Functions

Number, Aggregate, Character, Conversion and Date Functions - SQL Answers with Expected Outputs (MySQL 8.x)

Name: _____ Roll No: _____ Section: _____

The lab sheet does not give a schema, so the sample tables below are used. With different records the syntax stays the same but output values change.

Setup (run once before Task 1)

```
CREATE DATABASE CollegeDB_Functions;
USE CollegeDB_Functions;

CREATE TABLE Employees (Emp_ID INT PRIMARY KEY, First_Name VARCHAR(50), Last_Name VARCHAR(50),
                        Salary INT, Joining_Date DATE);
INSERT INTO Employees VALUES
(101, 'John', 'Doe', 45000, '2022-01-15'),
(102, 'Alice', 'Smith', 52000, '2021-06-20'),
(103, 'Rahul', 'Kumar', 38000, '2023-03-10'),
(104, 'Priya', 'Sharma', 61000, '2020-11-05'),
(105, 'David', 'Brown', 47500, '2022-08-18');

CREATE TABLE Students (Student_ID INT PRIMARY KEY, Student_Name VARCHAR(50));
INSERT INTO Students VALUES (1, 'Aarav'), (2, 'Priya'), (3, 'Rahul'), (4, 'Ananya'), (5, 'Karan');
```

A. Number Functions

1. Demonstrate ABS() with a suitable example.

SQL Answer:

```
SELECT ABS(-25) AS result;
```

Expected Output:

result
25

Note: ABS() returns the absolute (positive) value of a number.

2. Demonstrate ROUND() with a suitable example.

SQL Answer:

```
SELECT ROUND(123.4567, 2) AS result;
```

Expected Output:

result
123.46

Note: ROUND() rounds a number to the given number of decimal places.

3. Demonstrate CEIL() and FLOOR().

SQL Answer:

```
SELECT CEIL(12.34) AS ceil_value, FLOOR(12.34) AS floor_value;
```

Expected Output:

ceil_value	floor_value
13	12

Note: CEIL() gives the smallest integer >= the value; FLOOR() gives the largest integer <= the value.

4. Demonstrate MOD() with a suitable example.

SQL Answer:

```
SELECT MOD(17, 5) AS remainder;
```

Expected Output:

remainder
2

Note: MOD() returns the remainder after division.

5. Calculate the rounded salary of employees.

SQL Answer:

```
SELECT Emp_ID, First_Name, ROUND(Salary, -3) AS Rounded_Salary
FROM Employees;
```

Expected Output:

Emp_ID	First_Name	Rounded_Salary
101	John	45000
102	Alice	52000
103	Rahul	38000
104	Priya	61000
105	David	48000

Note: ROUND(Salary, -3) rounds to the nearest thousand.

B. Aggregate Functions

6. Find the total number of students using COUNT().

SQL Answer:

```
SELECT COUNT(*) AS Total_Students FROM Students;
```

Expected Output:

Total_Students
5

7. Find the total salary of all employees using SUM().

SQL Answer:

```
SELECT SUM(Salary) AS Total_Salary FROM Employees;
```

Expected Output:

Total_Salary
243500

8. Find the average salary using AVG().

SQL Answer:

```
SELECT AVG(Salary) AS Average_Salary FROM Employees;
```

Expected Output:

Average_Salary
48700.0000

9. Find the highest salary using MAX().

SQL Answer:

```
SELECT MAX(Salary) AS Highest_Salary FROM Employees;
```

Expected Output:

Highest_Salary
61000

10. Find the lowest salary using MIN().

SQL Answer:

```
SELECT MIN(Salary) AS Lowest_Salary FROM Employees;
```

Expected Output:

Lowest_Salary
38000

C. Character Functions

11. Demonstrate UPPER() on student names.

SQL Answer:

```
SELECT Student_Name, UPPER(Student_Name) AS Upper_Name
FROM Students;
```

Expected Output:

Student_Name	Upper_Name
Aarav	AARAV
Priya	PRIYA
Rahul	RAHUL
Ananya	ANANYA
Karan	KARAN

12. Demonstrate LOWER() on student names.

SQL Answer:

```
SELECT Student_Name, LOWER(Student_Name) AS Lower_Name
FROM Students;
```

Expected Output:

Student_Name	Lower_Name
Aarav	aarav
Priya	priya
Rahul	rahul
Ananya	ananya
Karan	karan

13. Demonstrate LENGTH() on student names.

SQL Answer:

```
SELECT Student_Name, LENGTH(Student_Name) AS Name_Length
FROM Students;
```

Expected Output:

Student_Name	Name_Length
Aarav	5
Priya	5
Rahul	5
Ananya	6
Karan	5

Note: In MySQL, LENGTH() counts bytes. For these ASCII names it equals the number of characters.

14. Demonstrate CONCAT() using first and last names.

SQL Answer:

```
SELECT CONCAT(First_Name, ' ', Last_Name) AS Full_Name
FROM Employees;
```

Expected Output:

Full_Name
John Doe
Alice Smith
Rahul Kumar
Priya Sharma
David Brown

15. Demonstrate SUBSTRING()/SUBSTR() with a suitable example.

SQL Answer:

```
SELECT Student_Name, SUBSTRING(Student_Name, 1, 3) AS Short_Name
FROM Students;
```

Expected Output:

Student_Name	Short_Name
Aarav	Aar
Priya	Pri
Rahul	Rah
Ananya	Ana
Karan	Kar

Note: SUBSTR() is an alias of SUBSTRING() in MySQL.

D. Conversion Functions

16. Convert a number into a character/string value.

SQL Answer:

```
SELECT CAST(12345 AS CHAR) AS String_Value;
```

Expected Output:

String_Value
12345

17. Convert a string into a numeric value.

SQL Answer:

```
SELECT CAST('12345' AS UNSIGNED) AS Numeric_Value;
```

Expected Output:

Numeric_Value
12345

18. Convert a date value into a different format.

SQL Answer:

```
SELECT DATE_FORMAT('2026-09-30', '%d-%m-%Y') AS Formatted_Date;
```

Expected Output:

Formatted_Date
30-09-2026

19. Demonstrate CAST() with a suitable example.

SQL Answer:

```
SELECT CAST('2026-09-30' AS DATE) AS Converted_Date;
```

Expected Output:

Converted_Date
2026-09-30

20. Demonstrate CONVERT() where supported by the DBMS.

SQL Answer:

```
SELECT CONVERT('2026-09-30', DATE) AS Converted_Date;
```

Expected Output:

Converted_Date
2026-09-30

E. Date Functions

21. Display the current date using a suitable date function.

SQL Answer:

```
SELECT CURDATE() AS `Current_Date`;
```

Expected Output:

Current_Date
2026-10-12

Note: The output shows the date on which the query is run. Backticks are used because Current_Date is a reserved word in MySQL.

22. Extract the year, month and day from a date.

SQL Answer:

```
SELECT YEAR('2026-09-30') AS `Year`,
       MONTH('2026-09-30') AS `Month`,
       DAY('2026-09-30') AS `Day`;
```

Expected Output:

Year	Month	Day
2026	9	30

23. Find the difference between two dates.

SQL Answer:

```
SELECT DATEDIFF('2026-09-30', '2026-01-01') AS Days_Difference;
```

Expected Output:

Days_Difference
272

24. Add a specified number of days to a date.

SQL Answer:

```
SELECT DATE_ADD('2026-09-30', INTERVAL 10 DAY) AS New_Date;
```

Expected Output:

New_Date
2026-10-10

25. Display employees' joining dates in a required date format.

SQL Answer:

```
SELECT Emp_ID, First_Name,
       DATE_FORMAT(Joining_Date, '%d/%m/%Y') AS Joining_Date
FROM Employees;
```

Expected Output:

Emp_ID	First_Name	Joining_Date
101	John	15/01/2022
102	Alice	20/06/2021
103	Rahul	10/03/2023
104	Priya	05/11/2020
105	David	18/08/2022

Note: DATE_FORMAT() changes only how the date is displayed, not the stored value.